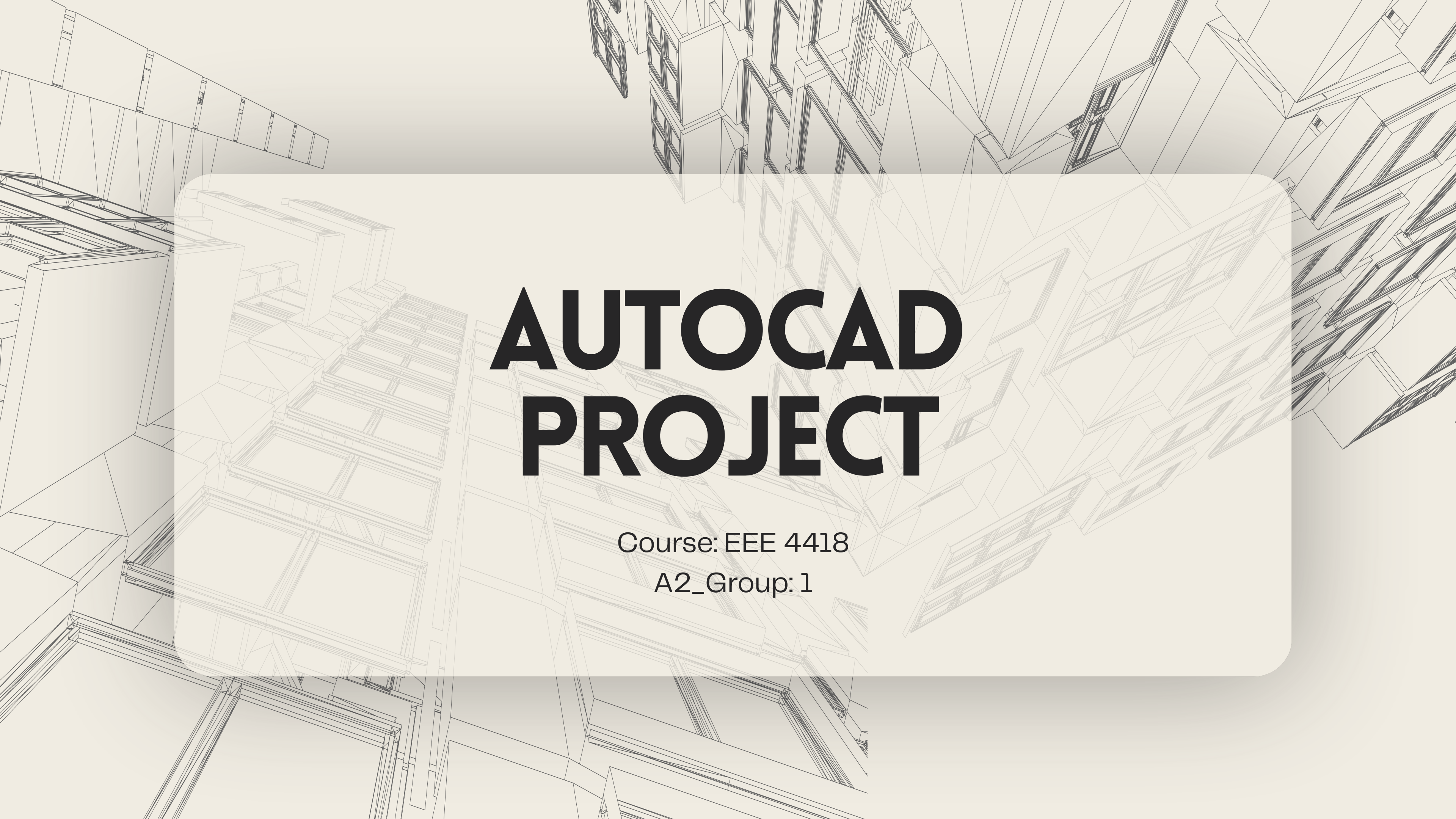


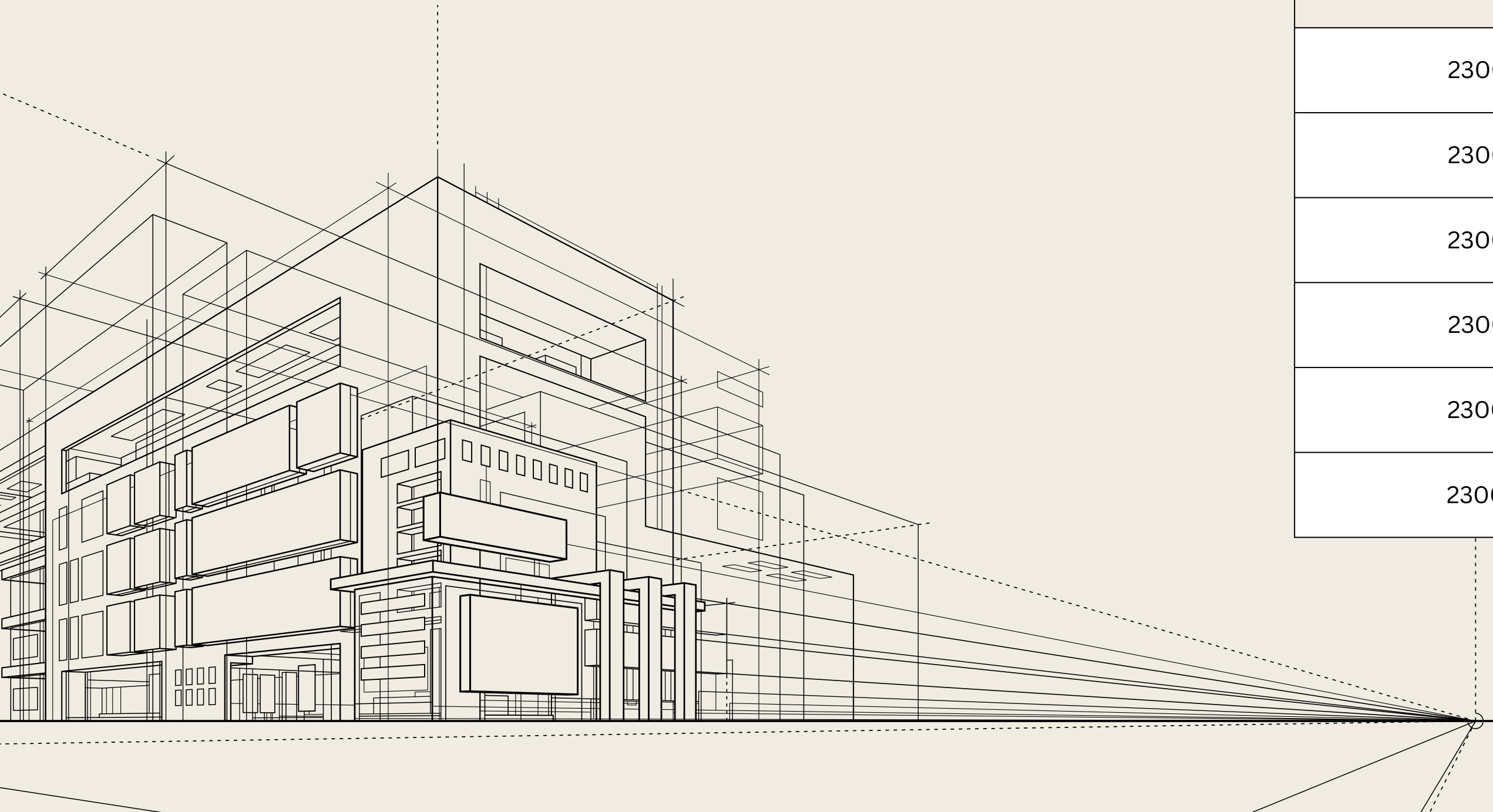


# AUTOCAD PROJECT

Course: EEE 4418

A2\_Group: 1

# GROUP MEMBERS



| Student ID | Student Name            |
|------------|-------------------------|
| 230021114  | Nadman Ibne Mamun       |
| 230021118  | Md Farhan Abrar Hossain |
| 230021128  | Tahmid Azim Ifaz        |
| 230021132  | Abdullah Al Muhit       |
| 230021134  | Samin Mohtasin          |
| 230021146  | Saif Ahmed              |



# INTRODUCTION

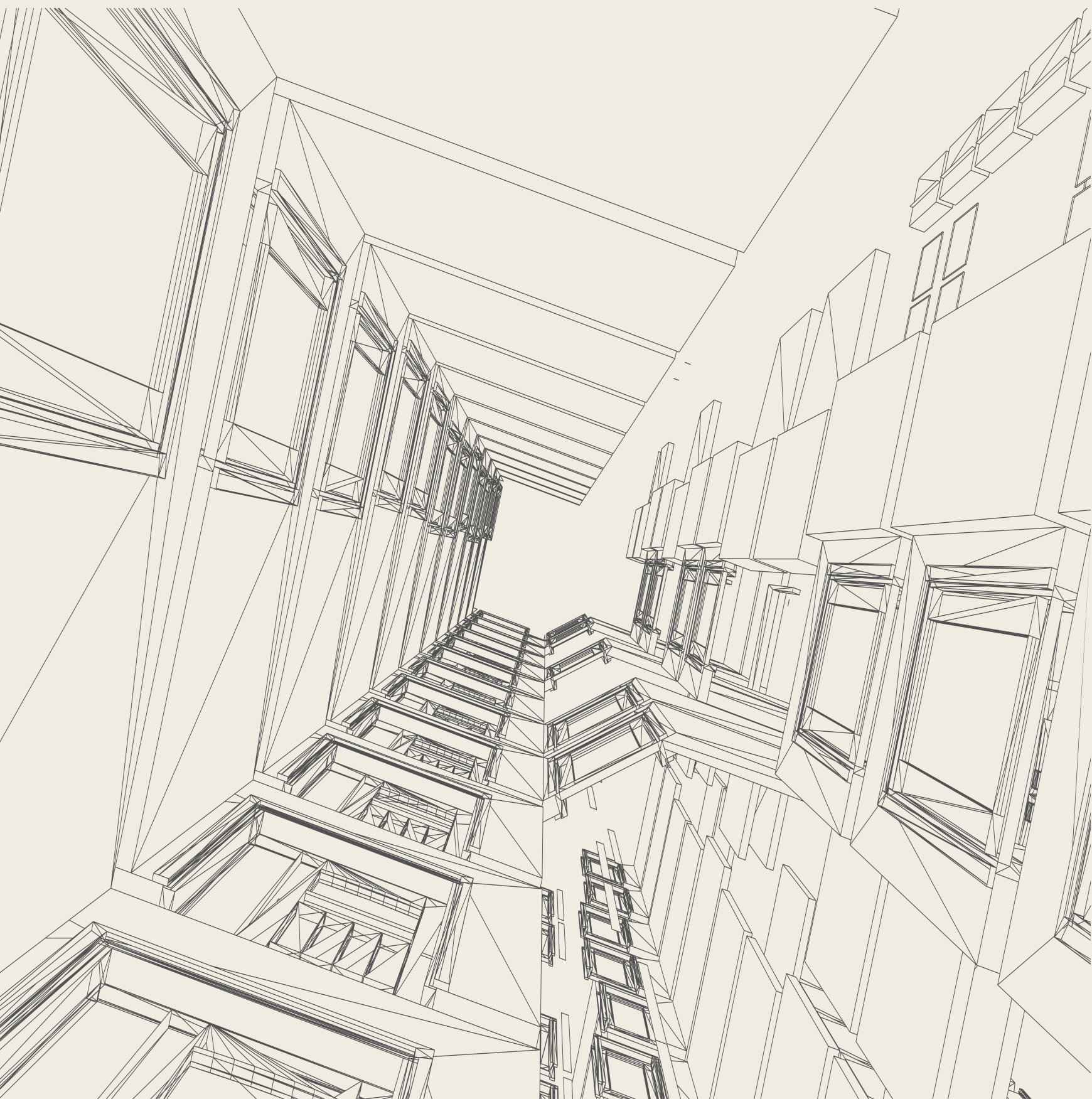
This project presents a complete residential building design developed using AutoCAD 2024, featuring carefully planned layouts for the ground floor, typical residential floors, garage, and rooftop. The typical residential floor consists of multiple well-planned units arranged around a central staircase and circulation area. Each unit is designed to provide comfort and functionality, incorporating spacious bedrooms, a kitchen, two toilets, and a combined living and dining space with proper access and ventilation.

The ground floor is primarily designed for parking and essential building services. It provides dedicated car parking spaces on both sides, along with a generator room, guard room, service/toilet facilities, and a centrally positioned staircase and platform for convenient vertical circulation. The layout ensures efficient use of the available space while maintaining clear movement and accessibility.

The rooftop is designed as a functional open space with designated areas for recreation and utility purposes. It includes a rooftop recreational area, a dedicated rooftop space, and water tanks positioned to support the building's water supply system. The arrangement provides both practical utility and an open environment for residents.

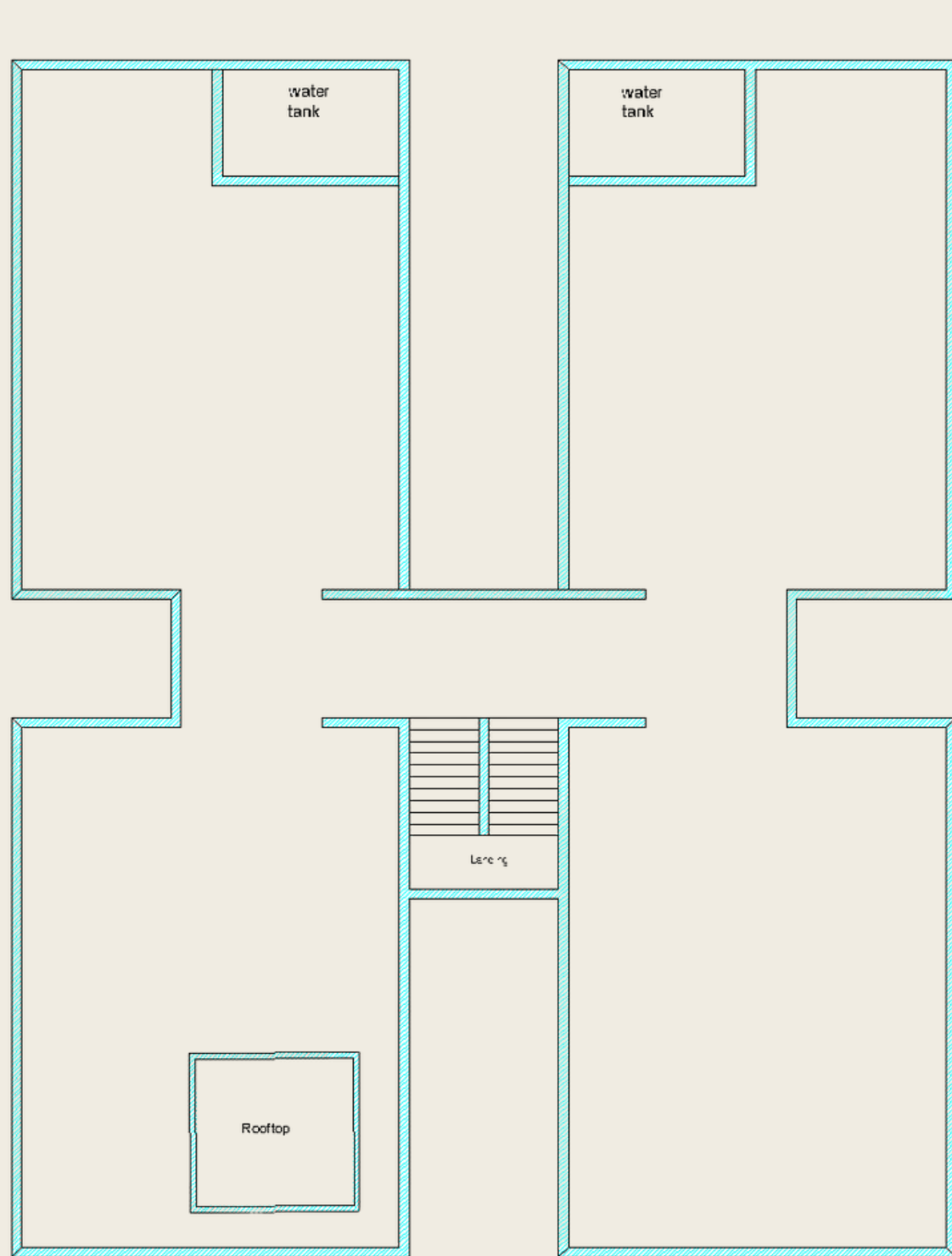
Overall, the project demonstrates a systematic approach to residential building planning, with emphasis on space utilization, functionality, accessibility, and resident comfort. The detailed AutoCAD drawings provide a clear representation of the architectural arrangement and essential facilities of the proposed building.



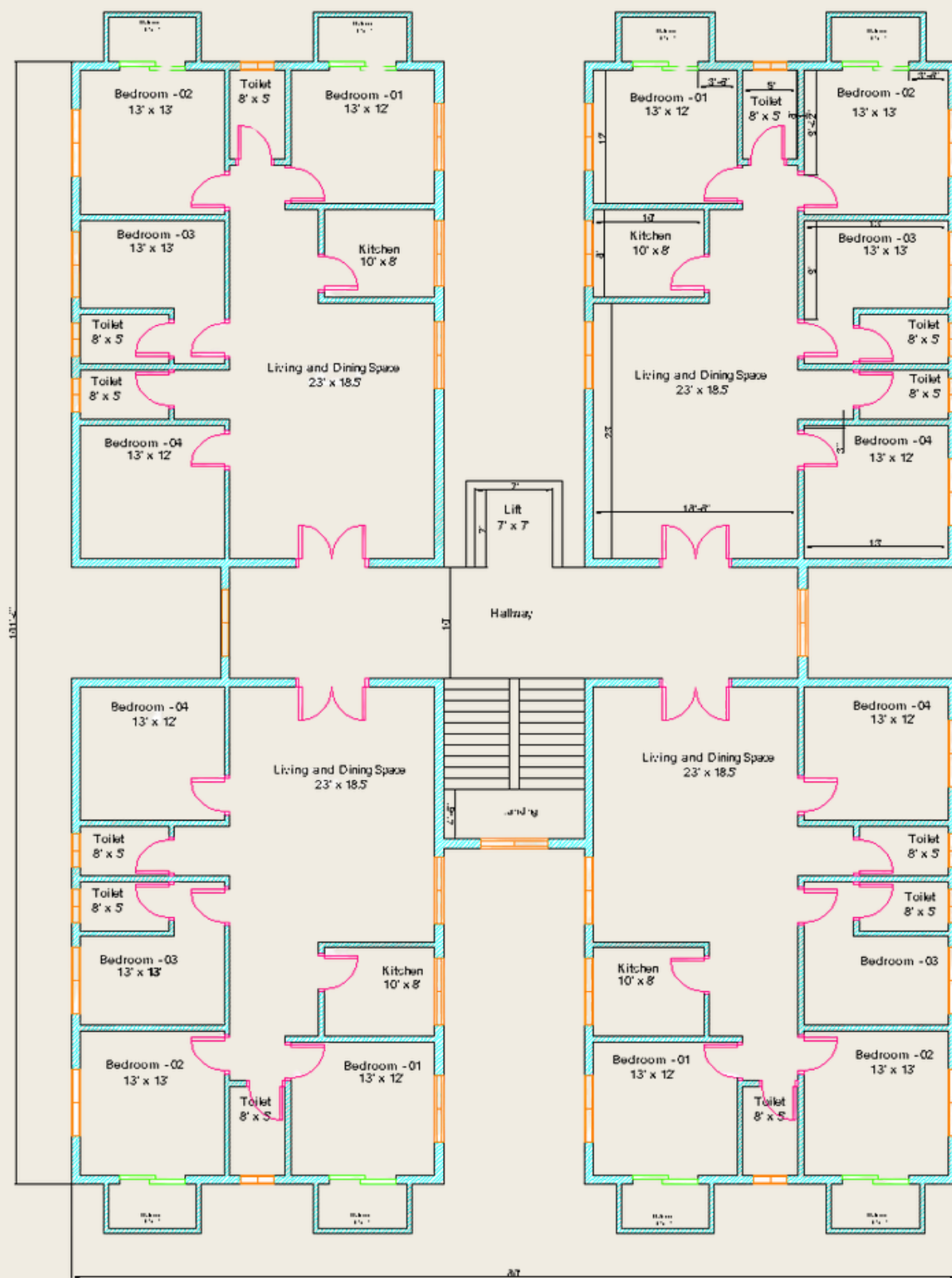


# OVERVIEW

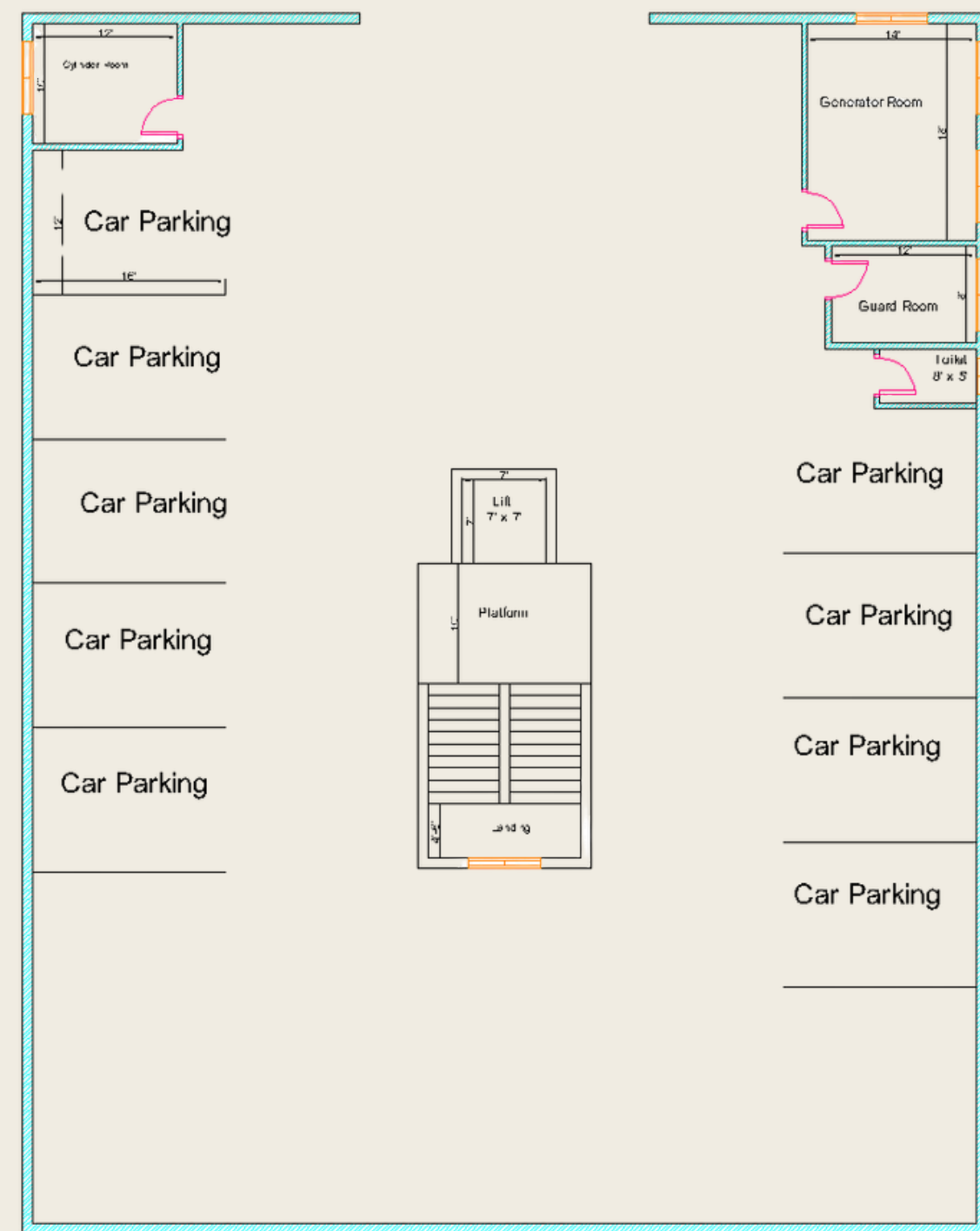




• ROOFTOP



• FLOOR



• GARAGE



# LEGEND APPLIANCE AND CONDUIT

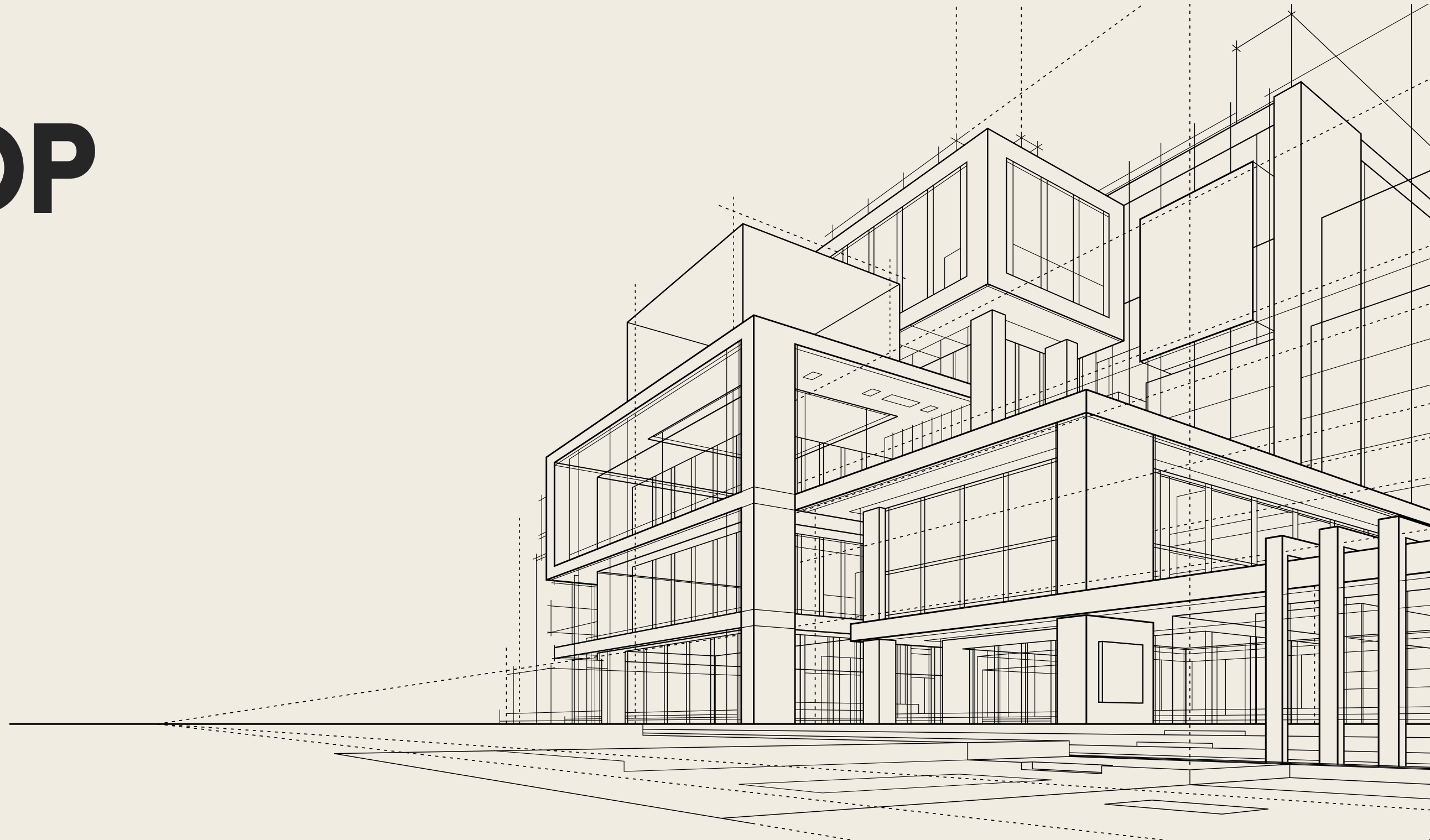


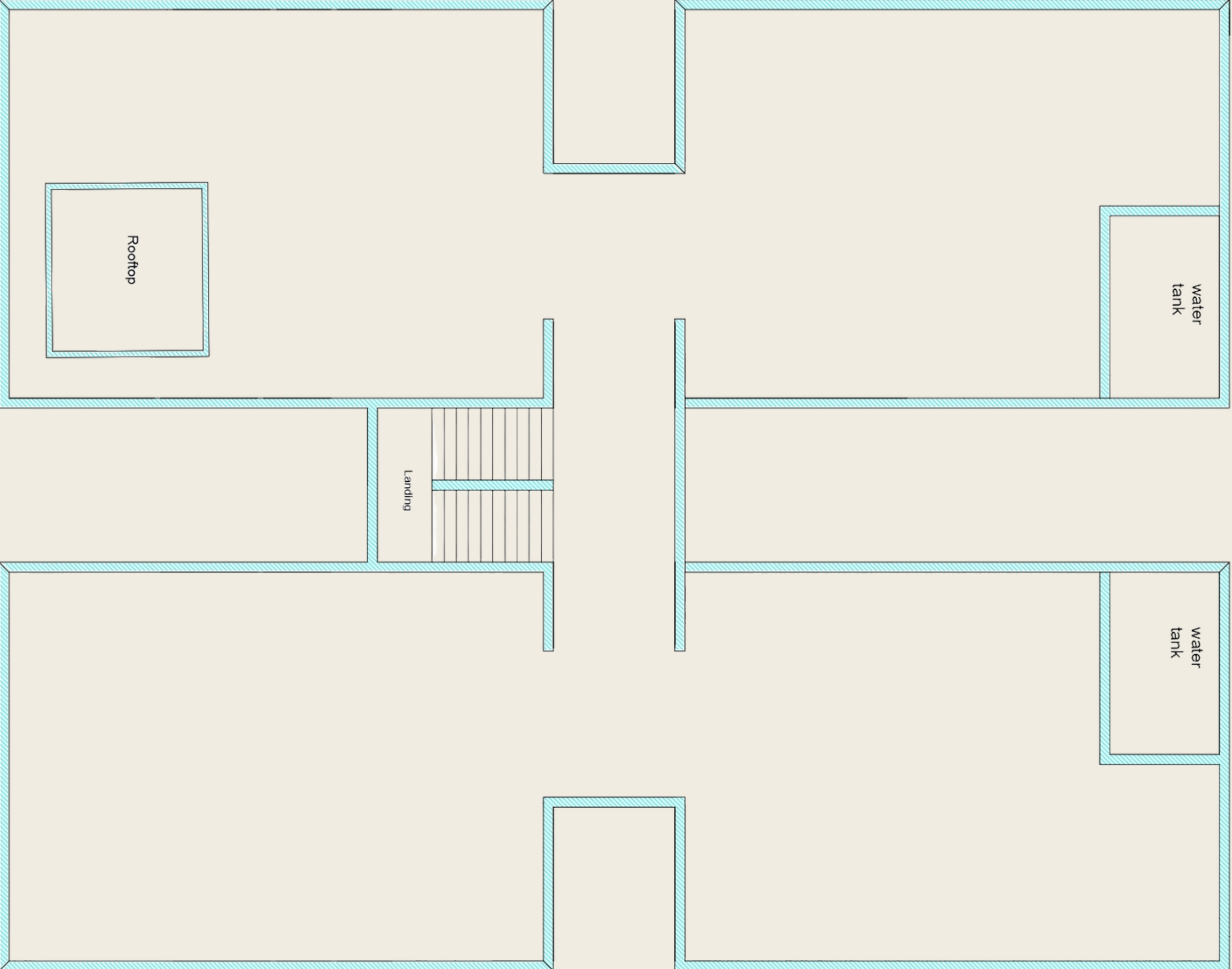
| LEGENDS                             |        |         |                                                                                       |
|-------------------------------------|--------|---------|---------------------------------------------------------------------------------------|
| DESCRIPTION                         | HEIGHT | CAPTION | SYMBOL                                                                                |
| Bracket-Mounted Light Bulb          |        | L       |    |
| Ceiling-Mounted Light Bulb          |        | CL      |    |
| Fan                                 |        | F       |    |
| Branch Distribution Board           |        | BDB     |    |
| Main Distribution Board             |        | MDB     |    |
| Emergency Branch Distribution Board |        | EBDB    |    |
| Switchboard                         |        | SB      |  |
| Emergency Switch Board              |        | ESB     |  |
| 2 Pin 5A Socket at SB level         |        |         |  |
| 3 Pin 20A Socket                    |        | Q       |  |
| Lift                                |        |         |                                                                                       |

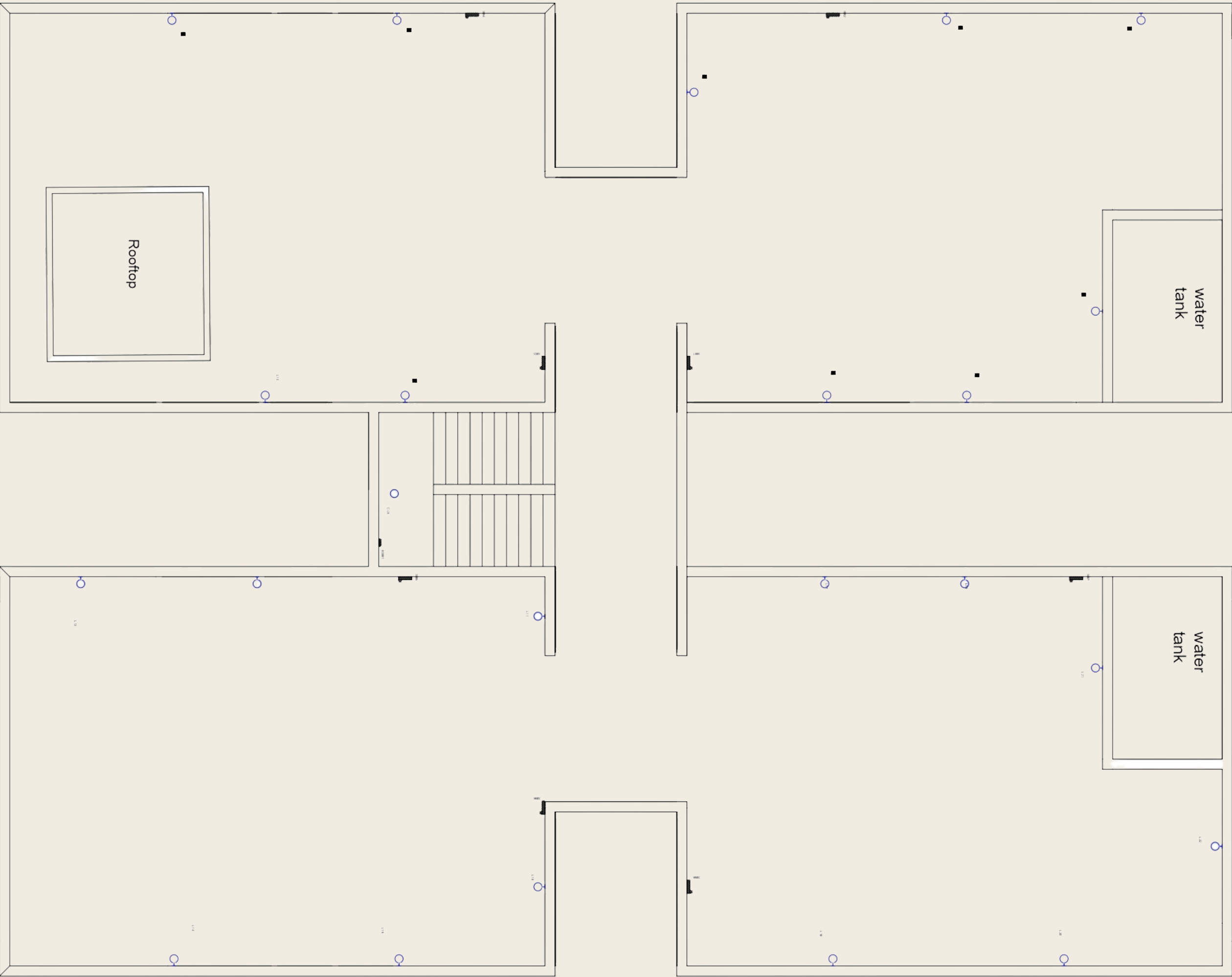
| LEGENDS FOR CONDUIT |                                                       |              |
|---------------------|-------------------------------------------------------|--------------|
| SYMBOL              | CONDUIT PROPERTIES                                    | CONDUIT SIZE |
| C1                  | 2 x 1.5 mm <sup>2</sup>                               | 3/4"         |
| C2                  | 4 x 1.5 mm <sup>2</sup>                               | 3/4"         |
| C3                  | 6 x 1.5 mm <sup>2</sup>                               | 3/4"         |
| C4                  | 8 x 1.5 mm <sup>2</sup>                               | 1"           |
| C5                  | 10 x 1.5 mm <sup>2</sup>                              | 1"           |
| C6                  | 12 x 1.5 mm <sup>2</sup>                              | 1"           |
| C7                  |                                                       |              |
| C9                  | 2 x 2.5 mm <sup>2</sup> and 1 x 4 mm <sup>2</sup> ECC | 1"           |



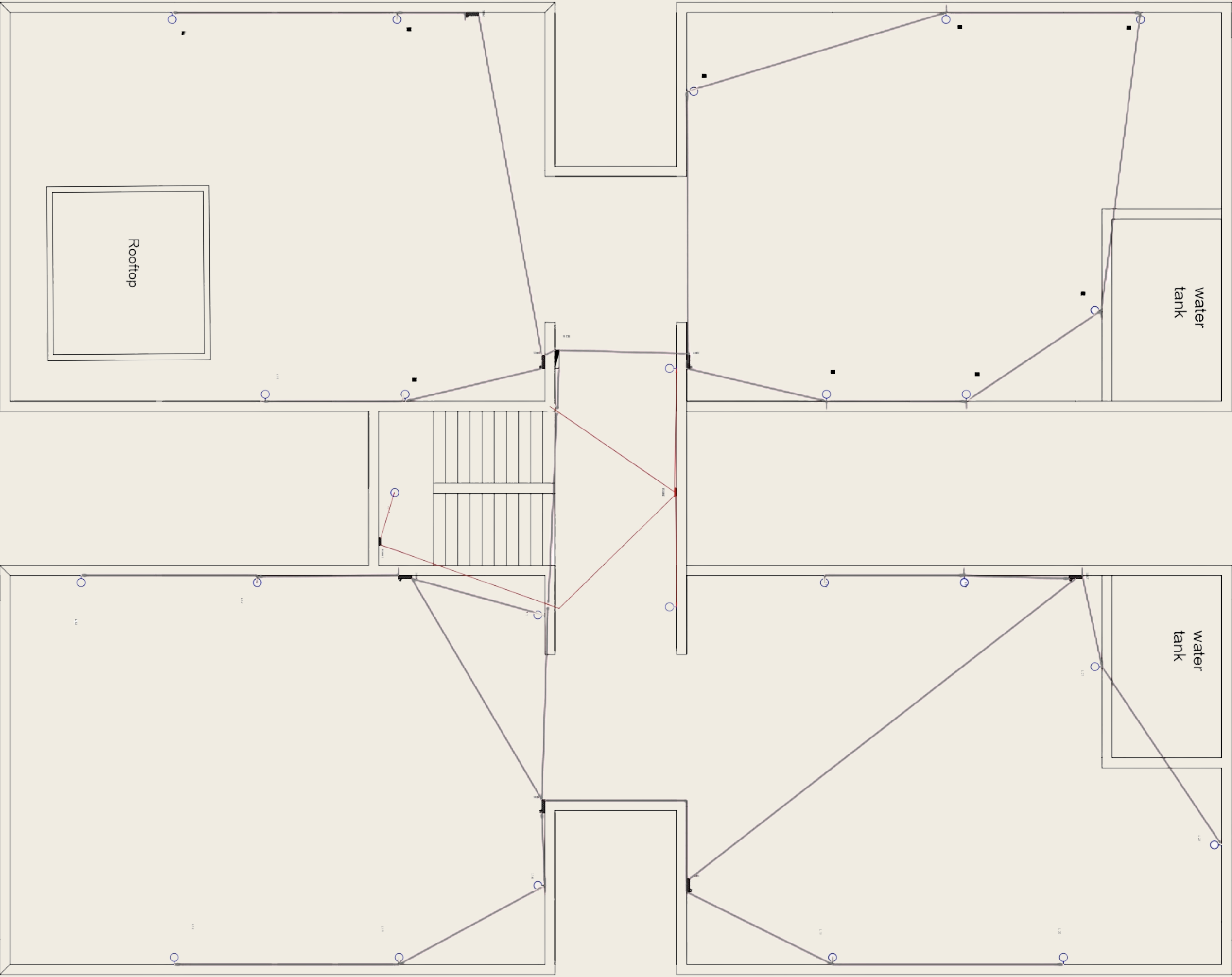
# ROOFTOP PLAN





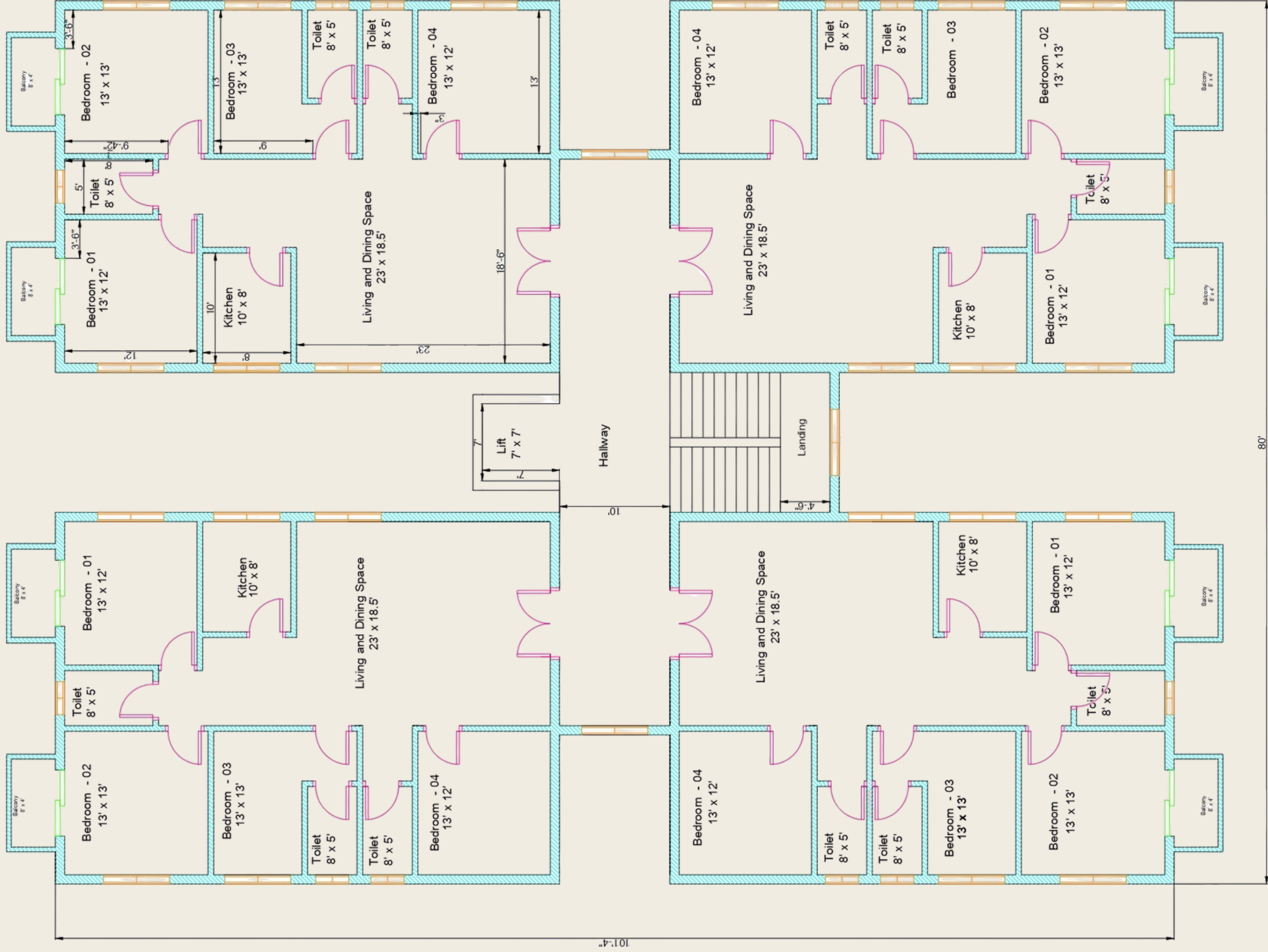


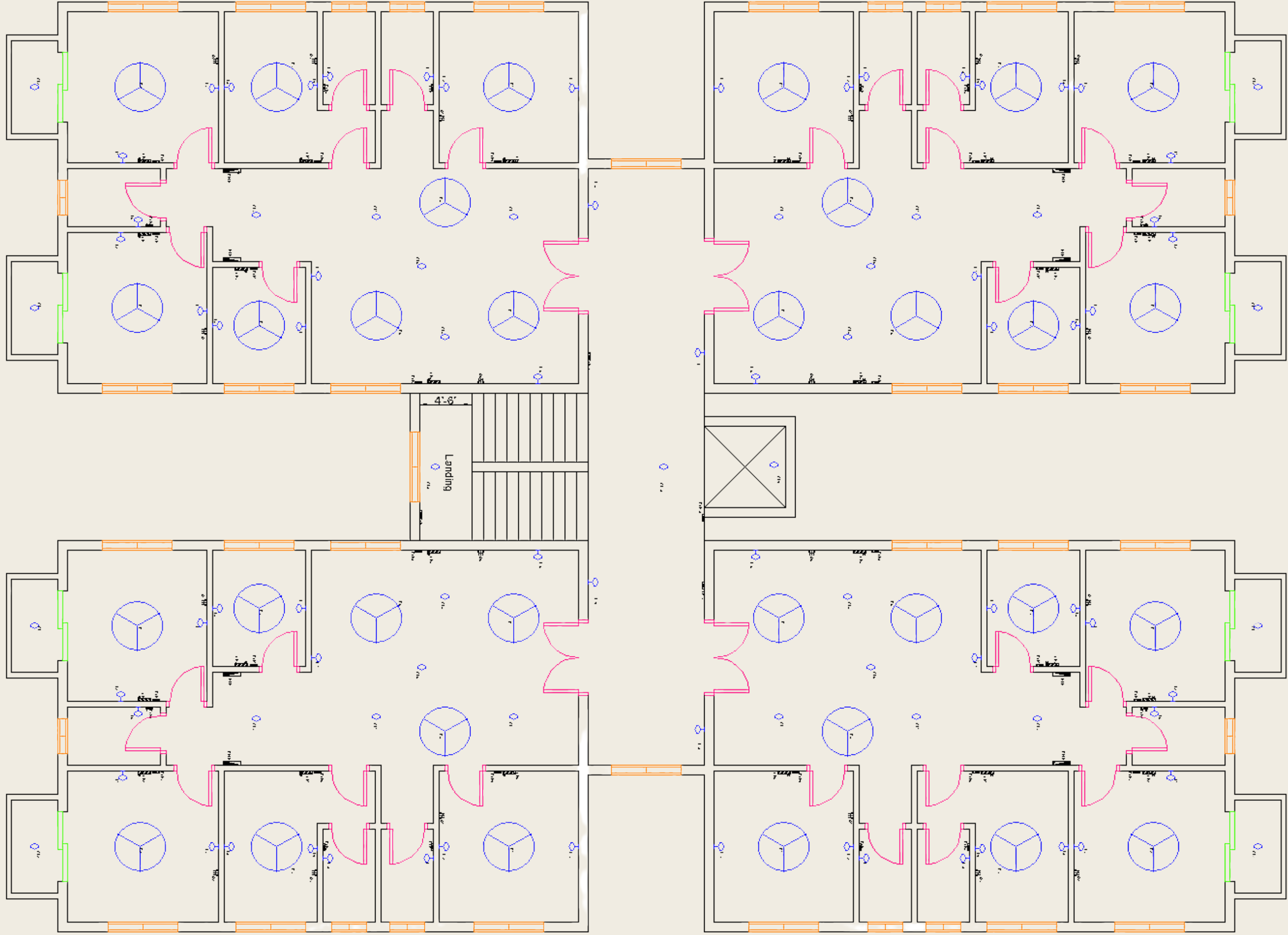




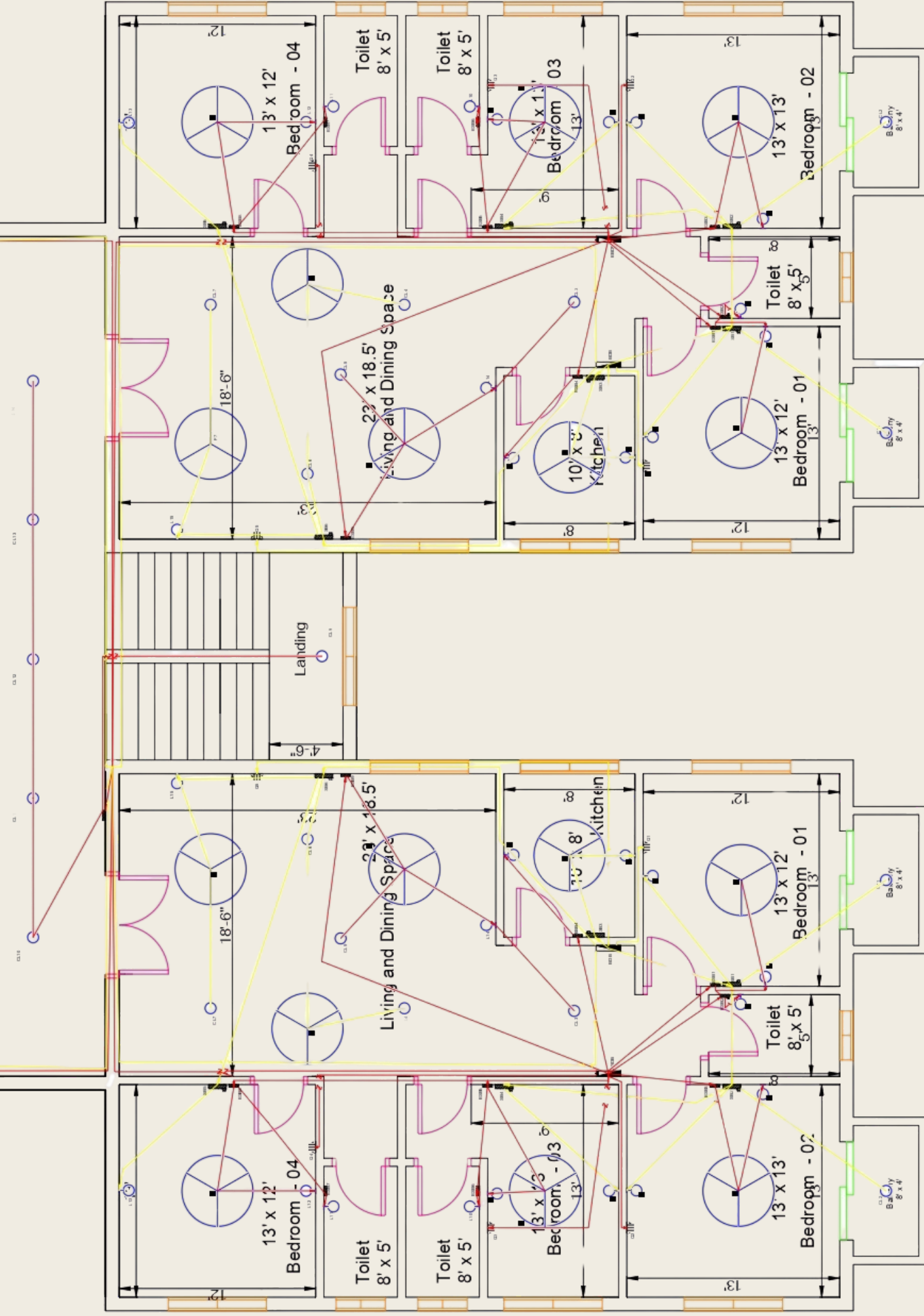
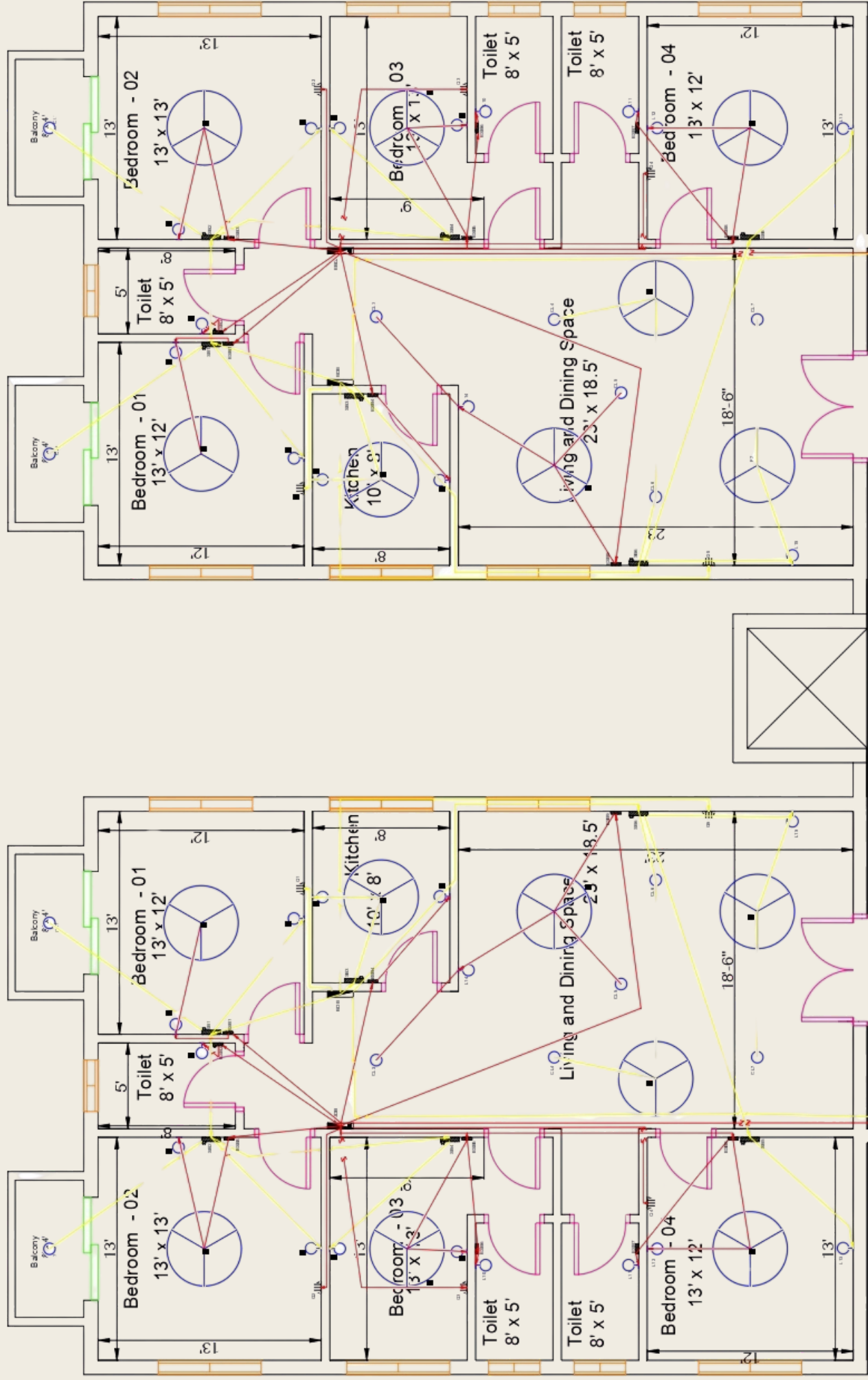
# FLOOR PLAN





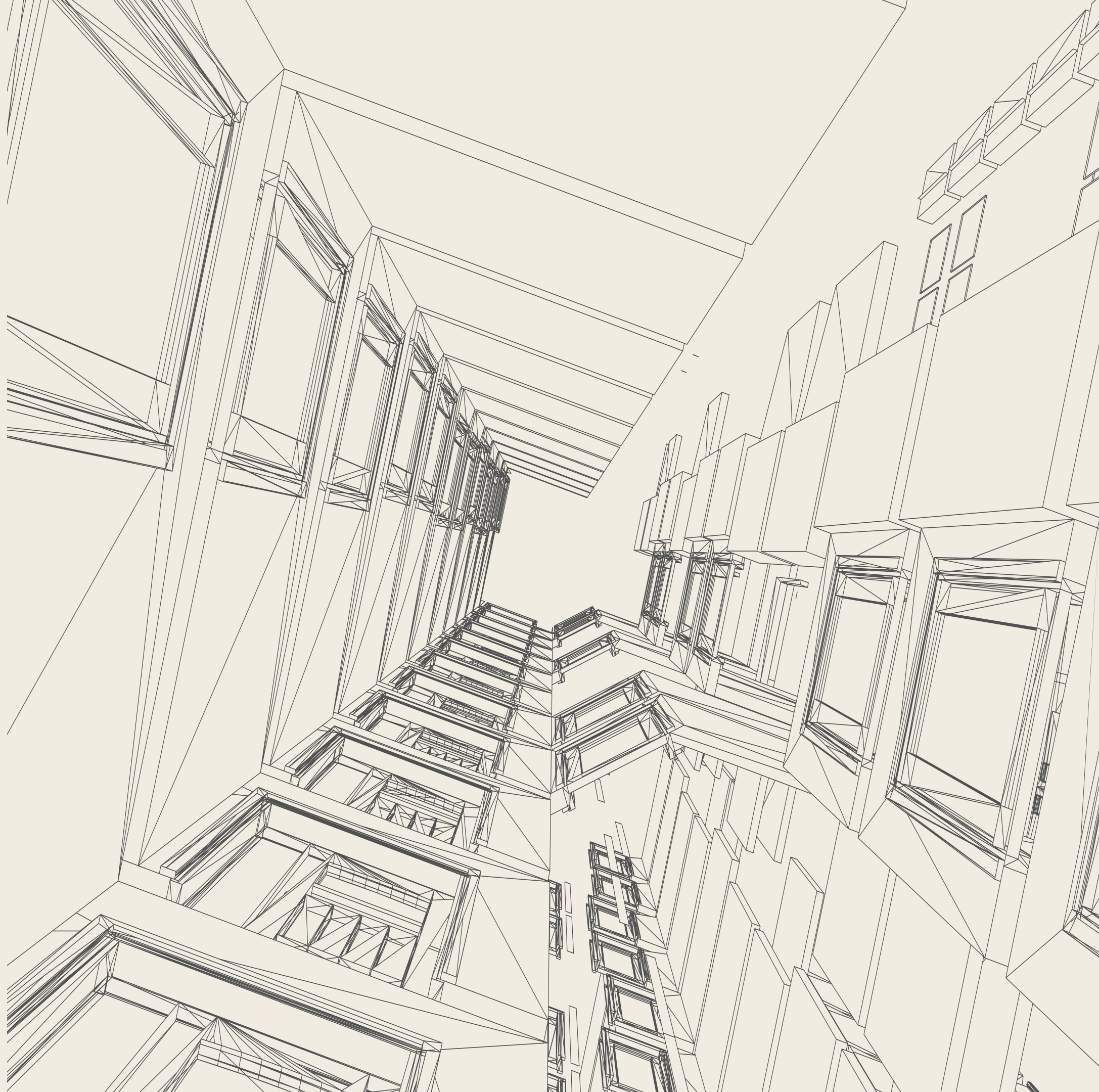


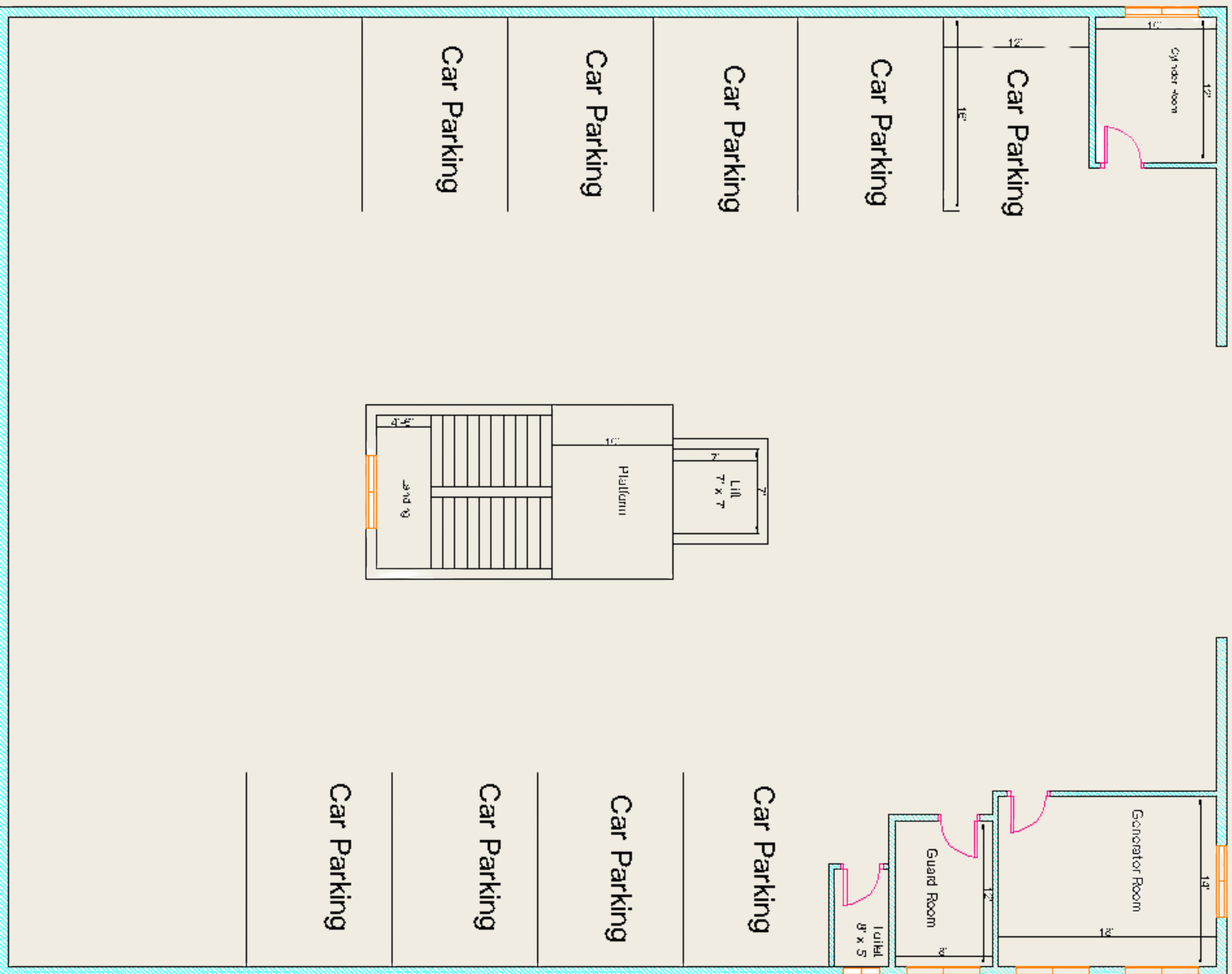


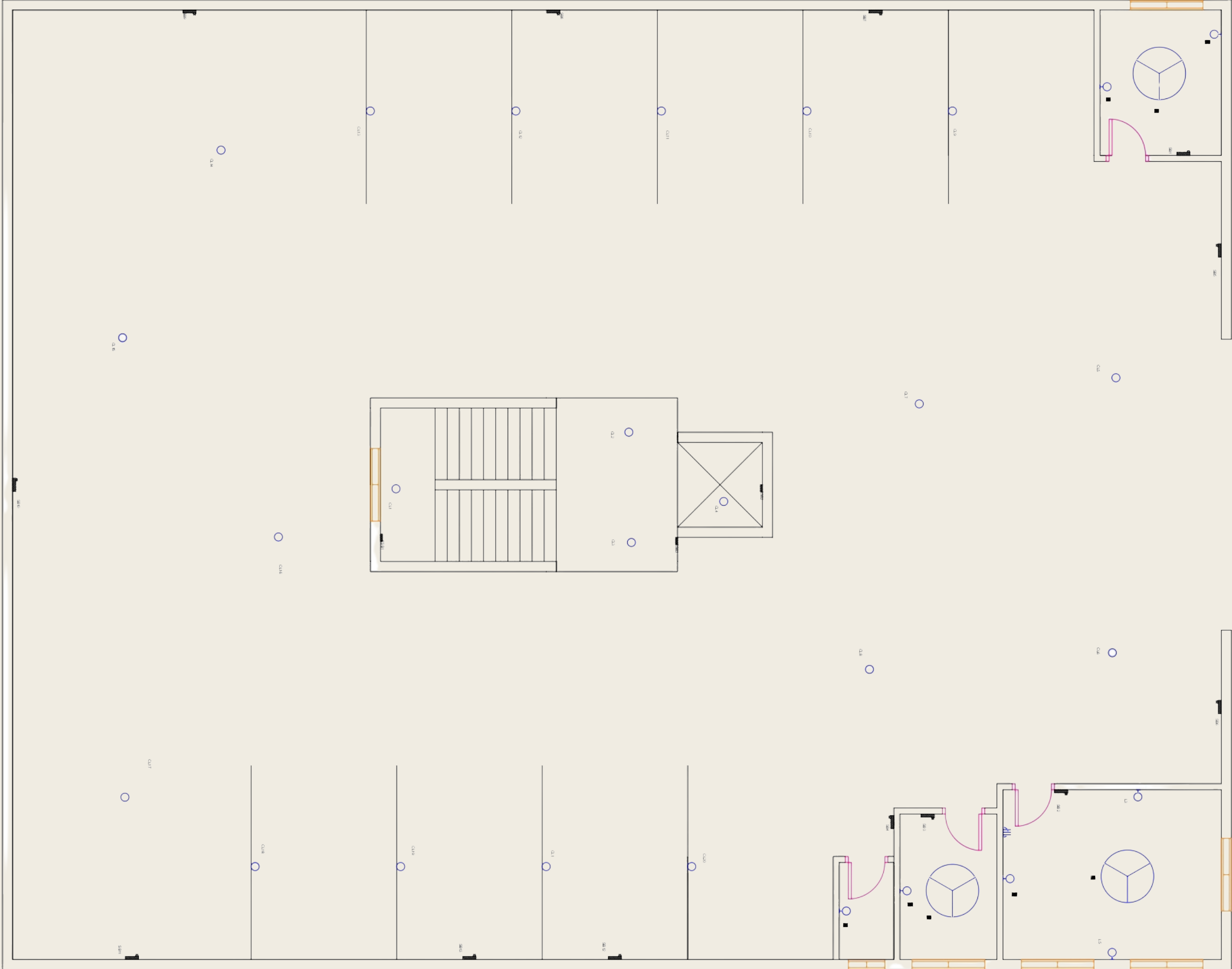




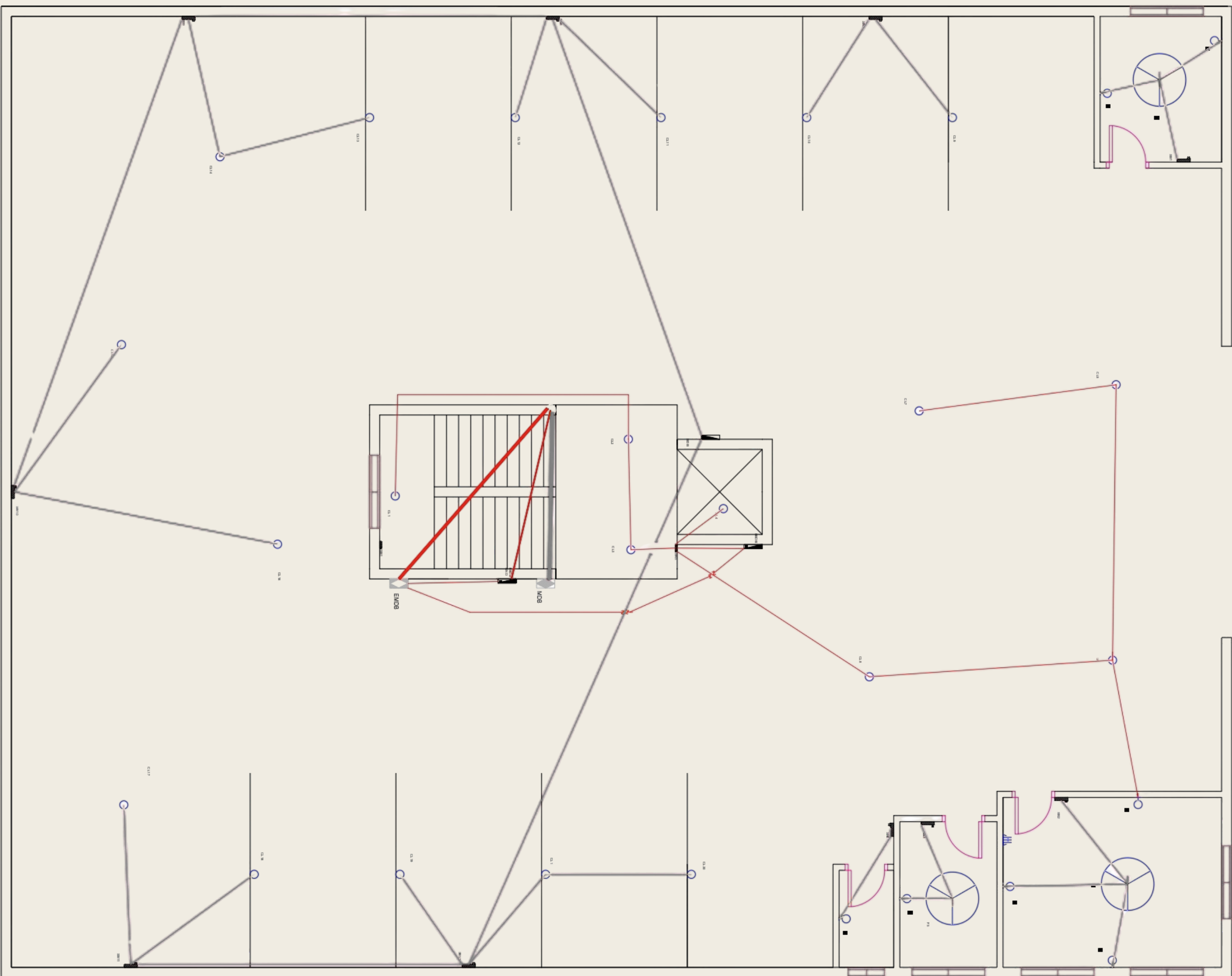
# GARAGE PLAN

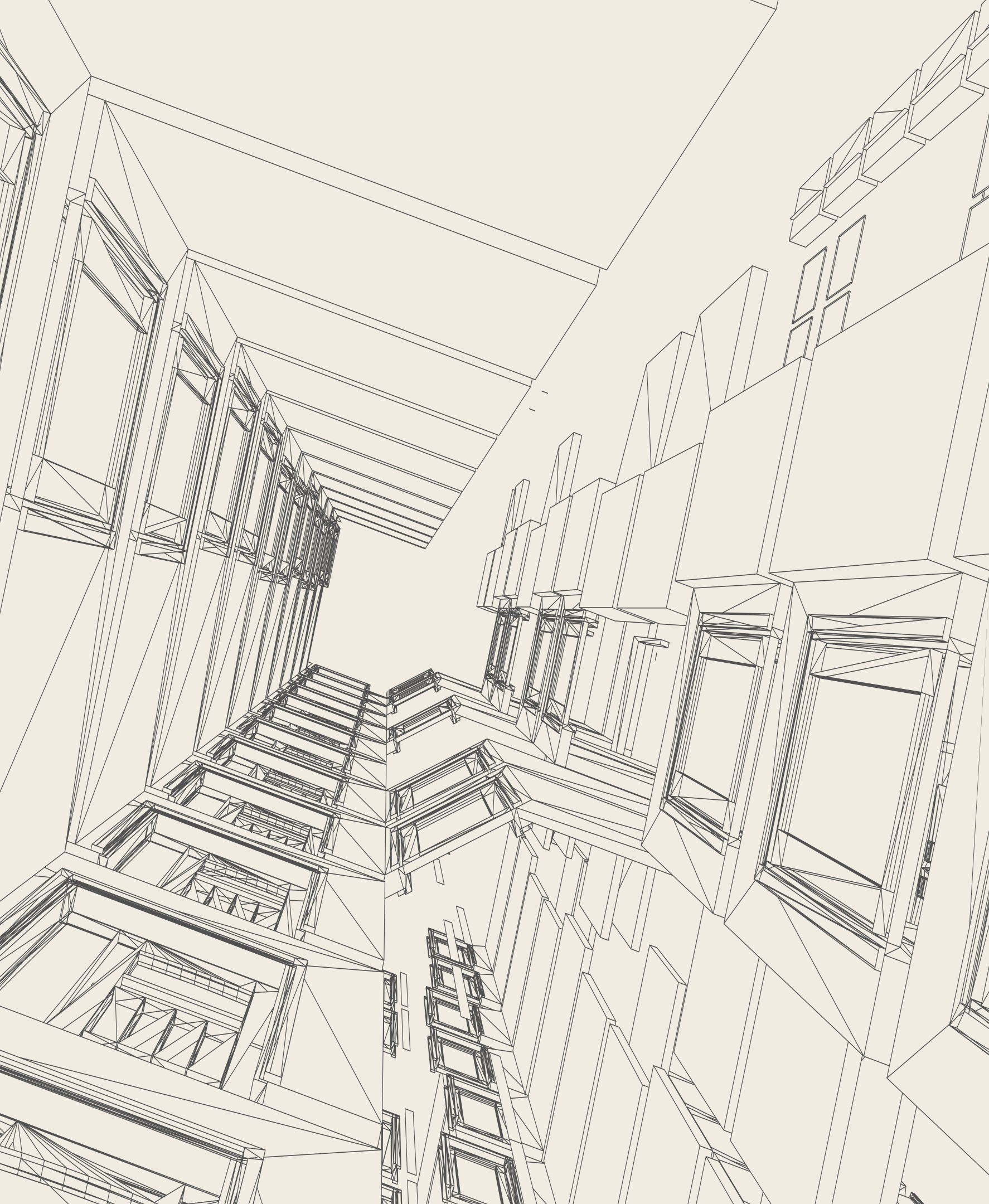






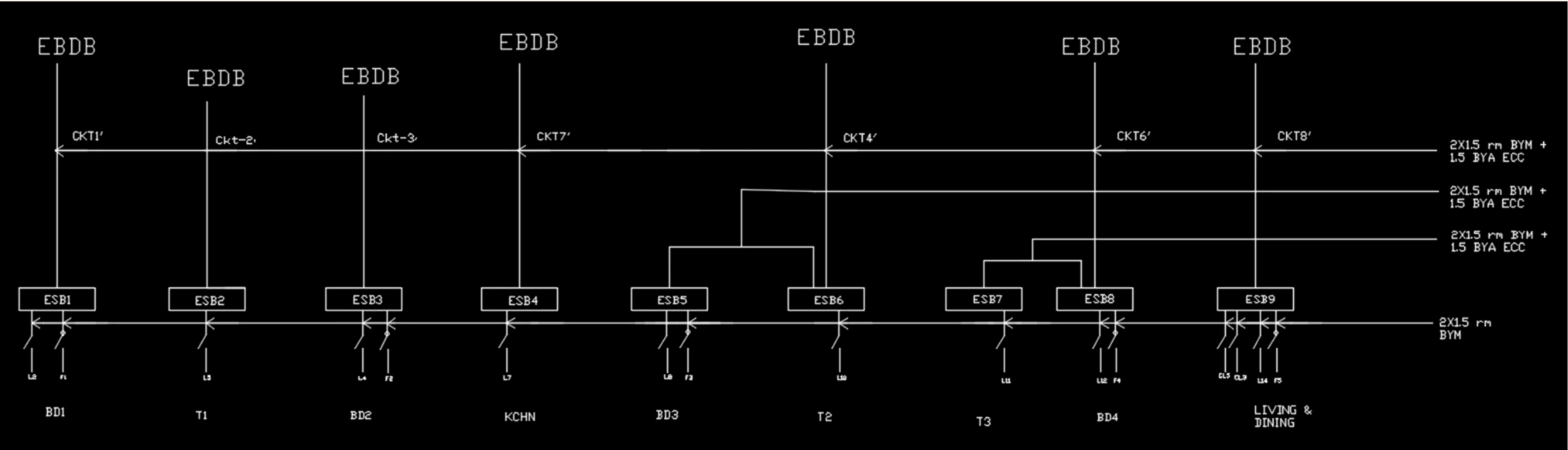






# SWITCHBOARD DIAGRAM

# SWITCHBOARD DIAGRAM (FLOOR)



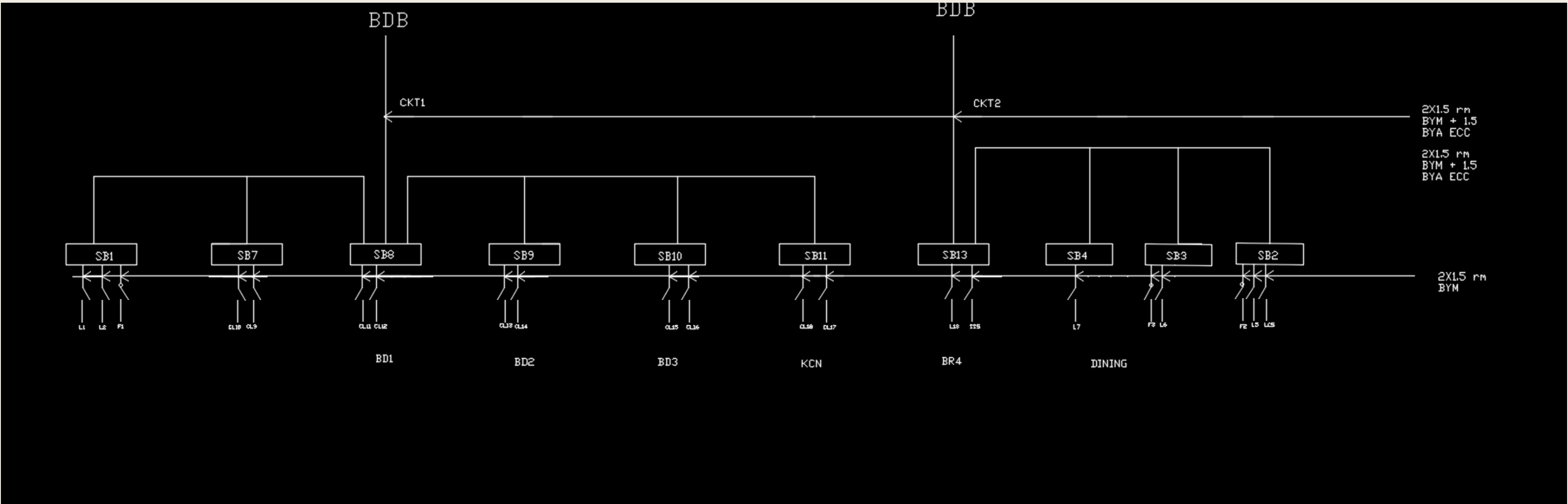


# BRANCH DISTRIBUTION DIAGRAM

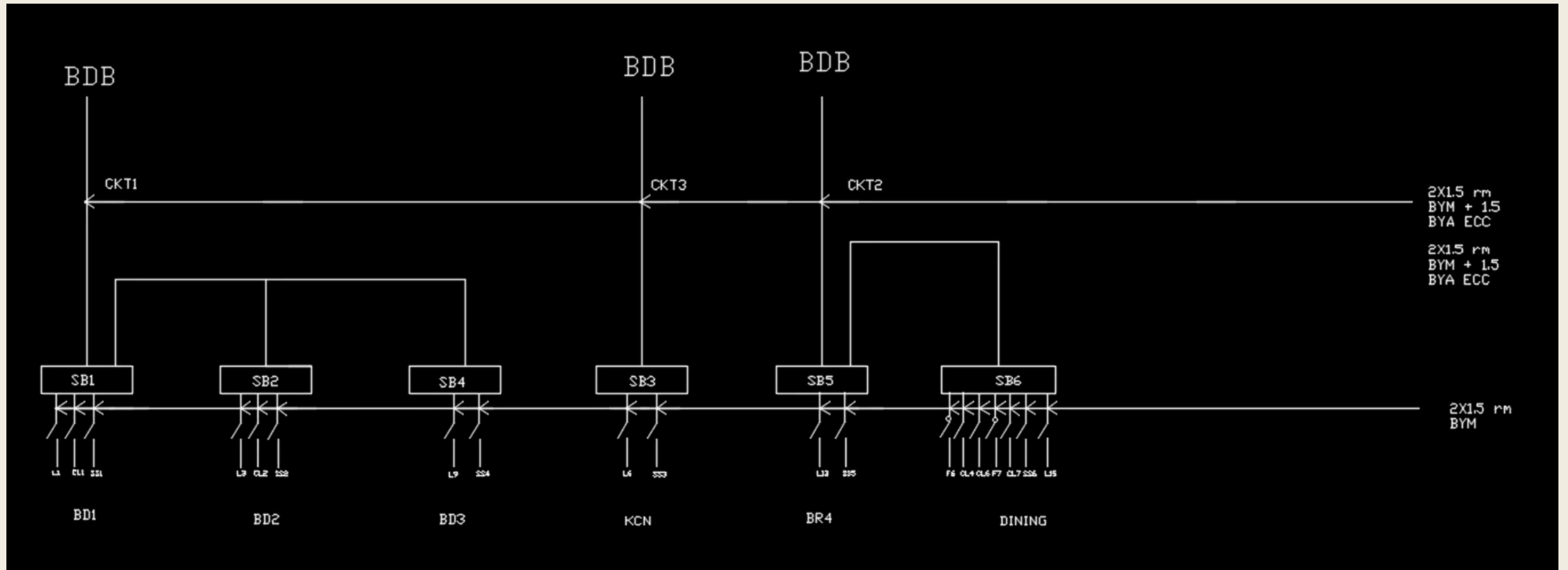




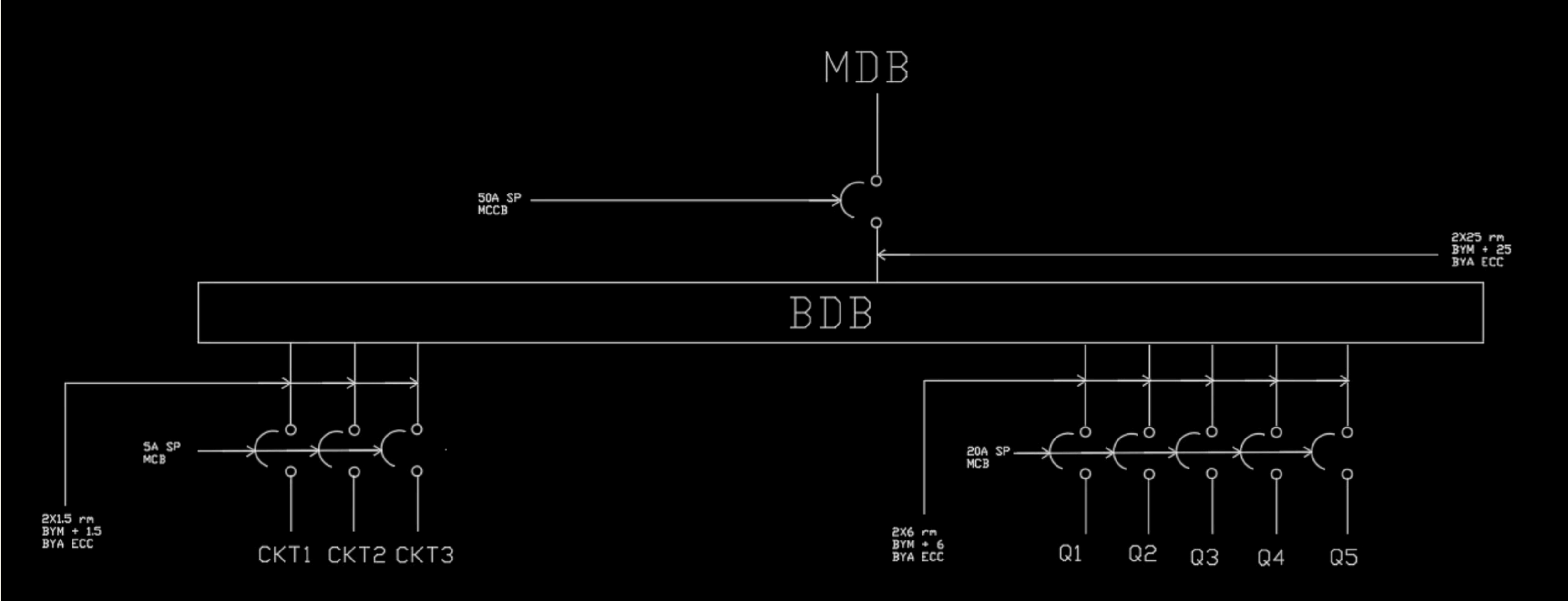
**BDB (GROUND)**



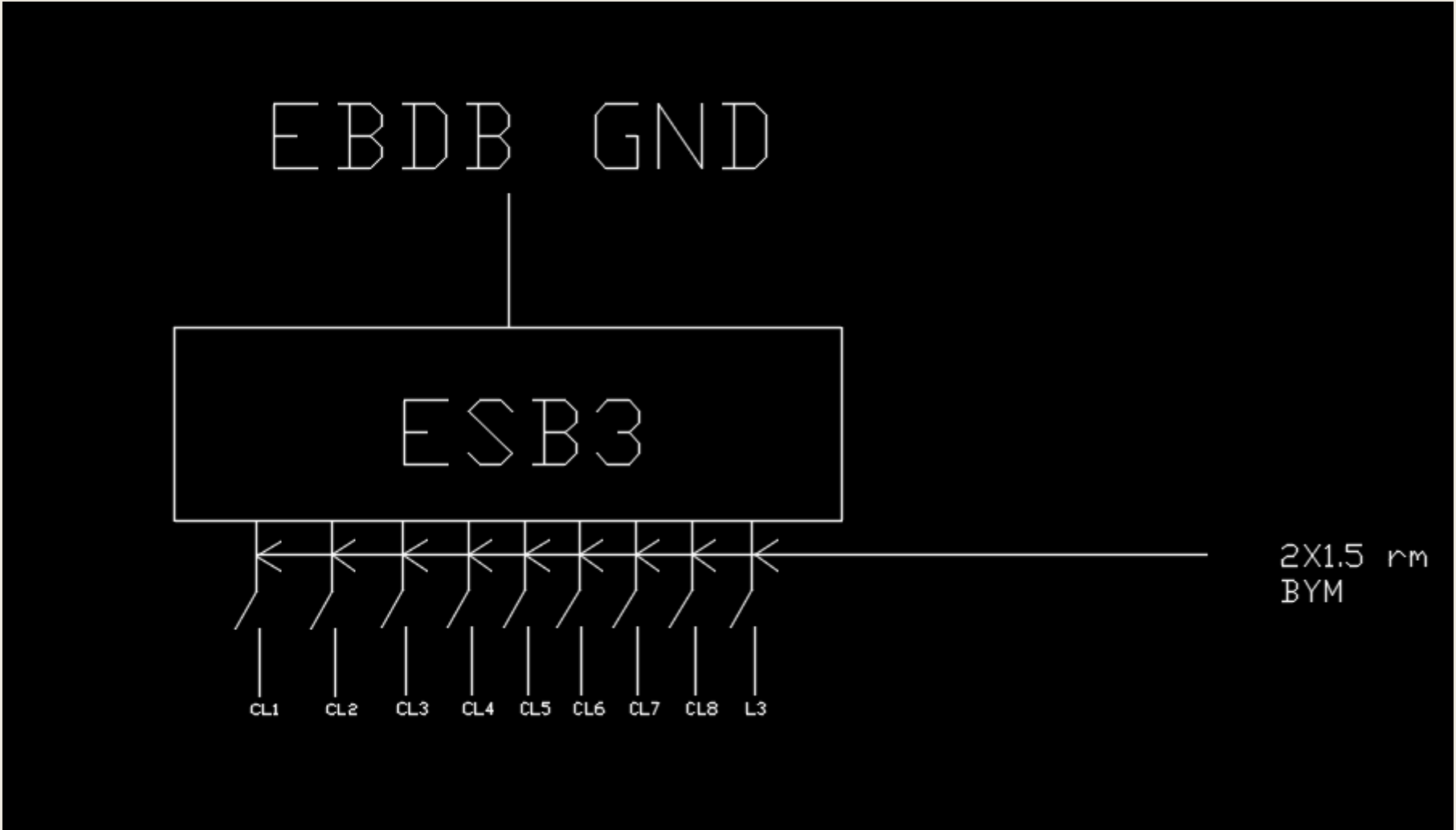
# BDB (FLOOR)



**BDB (ROOF)**

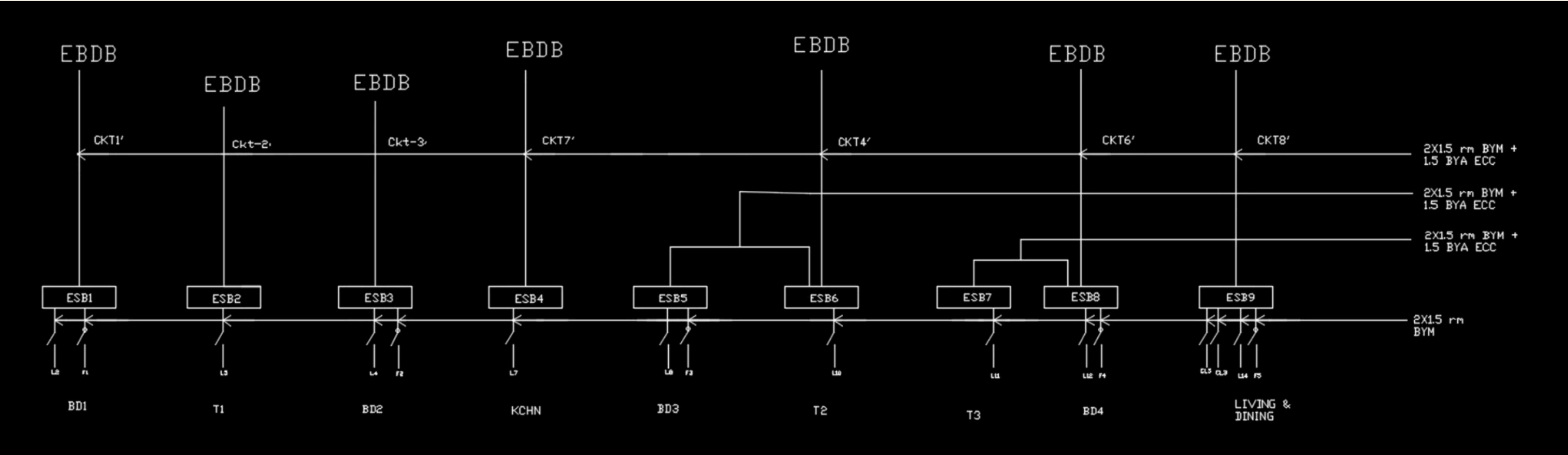


EBDB (GROUND)

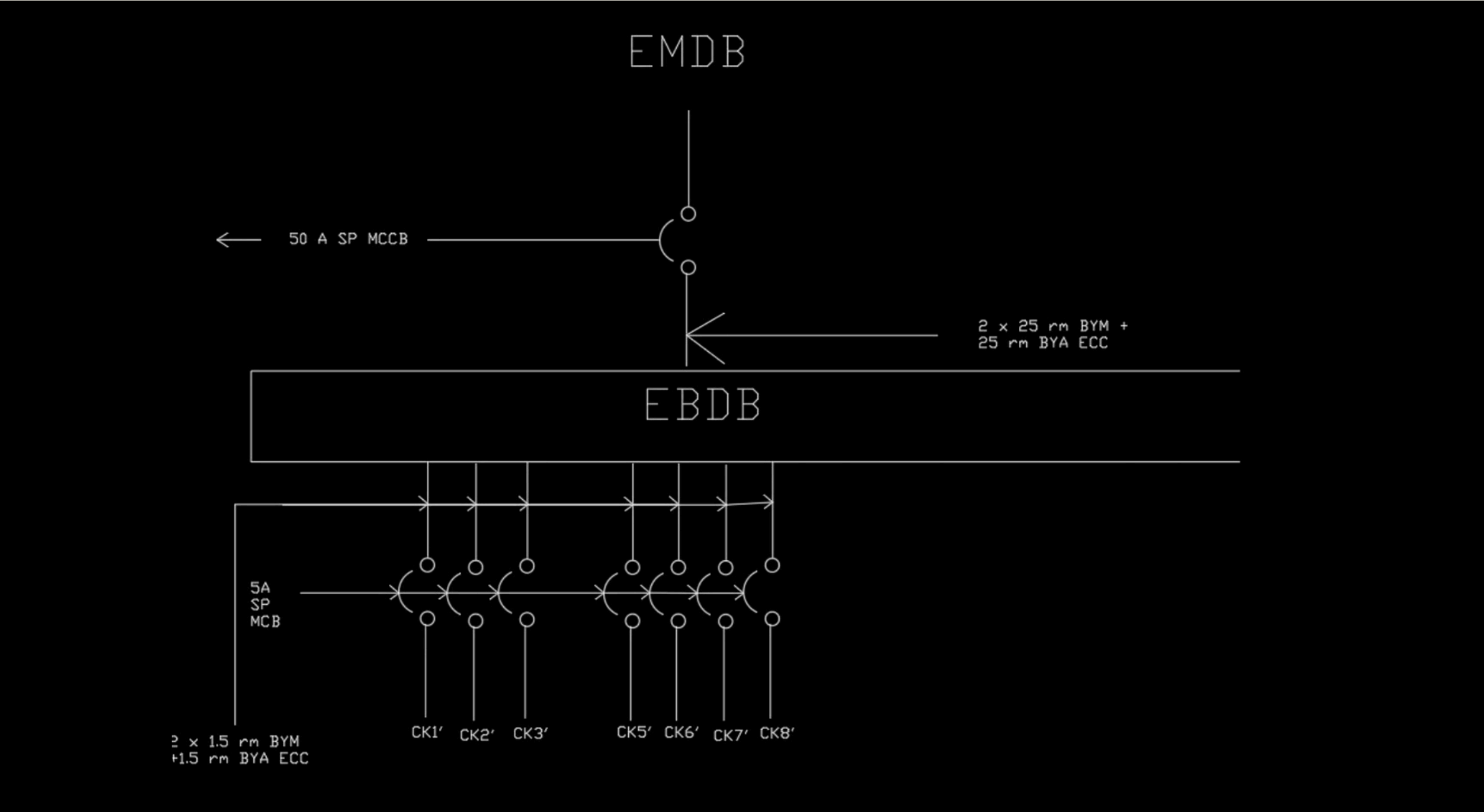




# EBDB (FLOOR)



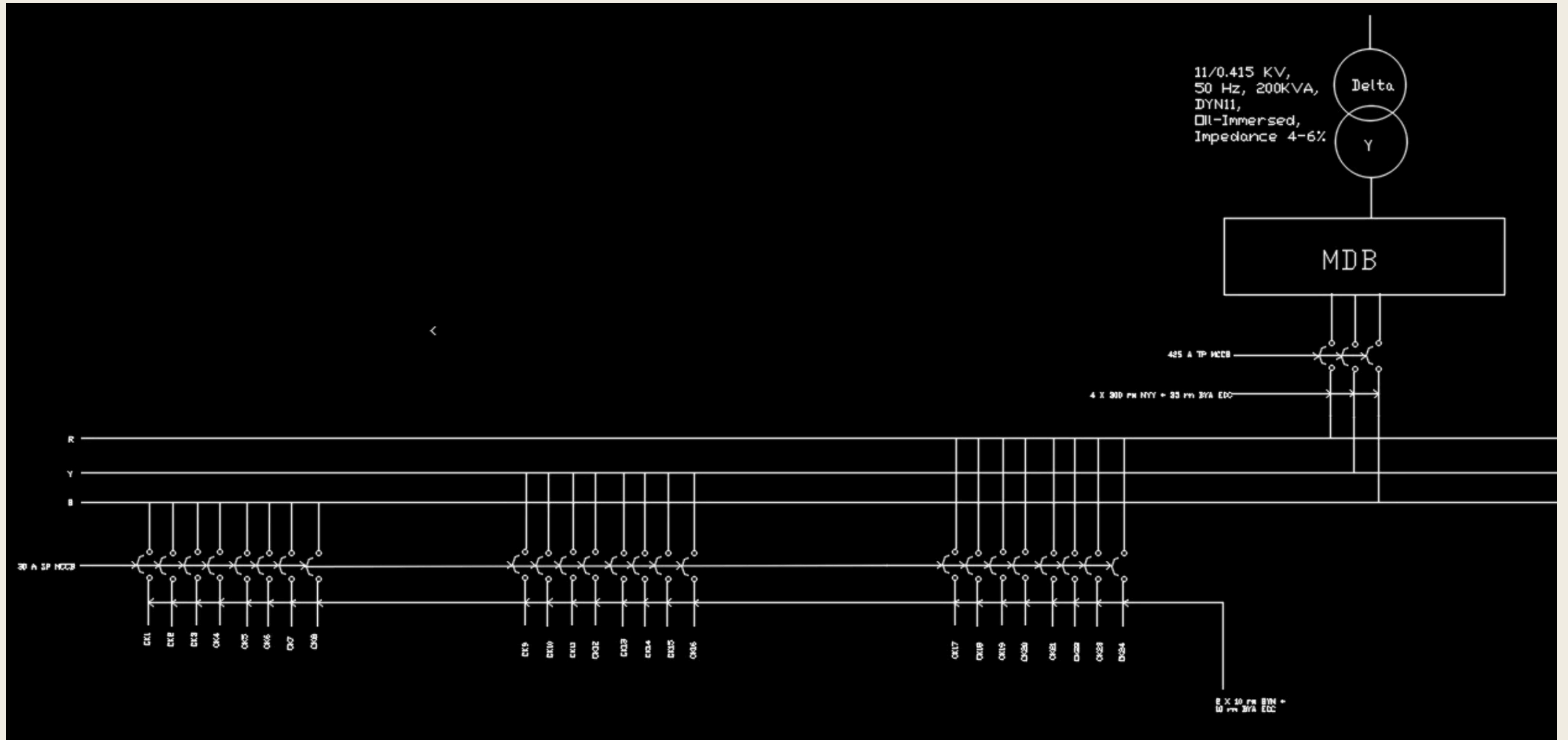
EBDB (ROOF)



# MAIN DISTRIBUTION BOARD DIAGRAM

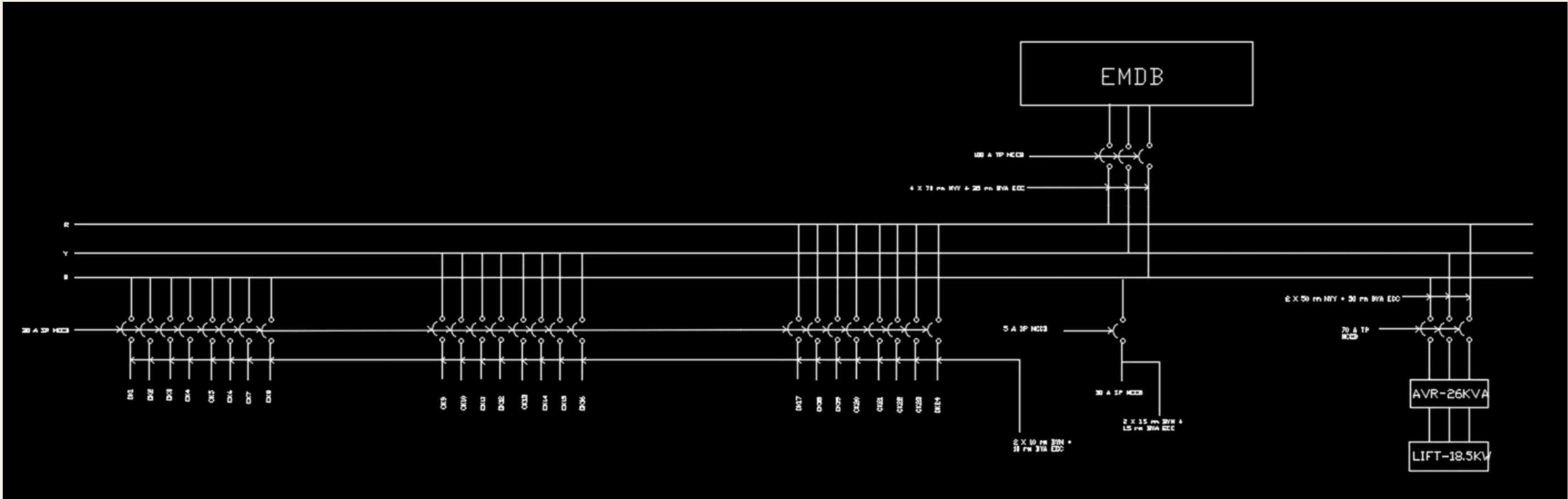


# MDB





# EMDB



# REFERENCES

| $\mathcal{A}$ | $\mathcal{B}$ | $\mathcal{C}$ | $\mathcal{D}$ | $\mathcal{E}$ | $\mathcal{F}$ |      | $\mathcal{G}$ | $\mathcal{H}$ | $\mathcal{I}$ |       | $\mathcal{J}$ |                  |
|---------------|---------------|---------------|---------------|---------------|---------------|------|---------------|---------------|---------------|-------|---------------|------------------|
|               |               |               |               |               | $a'$          | $b'$ |               |               | $a''$         | $b''$ | $a'''$        | $\mathcal{B}'''$ |
| 3/0.029       | 1.5           | 5             | 16            | 10            | 6             | 10   |               | 27            | 27            | 22    | 16            | 20               |
| 7/0.029       | 2.5           | 10            | 16            | 10            | 4             | 7    |               | 16            | 36            | 30    | 22            | 28               |
| 7/0.036       | 4             | 15            | 14            | 10            | 3             | 5    | 1             | 10            | 47            | 39    | 30            | 37               |
| 7/0.044       | 6             | 20            | 14            | 10            | 2             | 4    | 1             | 6.8           | 59            | 50    | 38            | 47               |
| 7/0.052       | 10            | 30            | 10            | 10            | 1             | 2    | 1.5           | 4             | 78            | 68    | 52            | 63               |
| 7/0.064       | 16            | 40            | 10            | 10            |               | 1    | 1.5           | 2.6           | 100           | 94    | 70            | 85               |
| 19/0.052      | 25            | 50            | 6             | 6             |               | 1    | 2             | 1.6           | 130           | 125   | 91            | 110              |
| 19/0.064      | 35            | 60            | 6             | 6             |               |      | 2             | 1.2           | 155           | 160   | 112           | 136              |
| 19/0.072      | 50            | 70            | 6             | 6             |               |      | 2             | 0.93          | 185           | 195   | 136           | 164              |
| 19/0.083      | 70            | 100           | 1/0           | 1/0           |               |      | 2             | 0.65          | 225           | 245   | 173           | 207              |
| 37/0.072      | 95            | 120           | 1/0           | 1/0           |               |      | 2.5           | 0.48          | 270           | 300   | 216           | 253              |
| 37/0.083      | 120           | 150           | 1/0           | 1/0           |               |      | 2.5           | 0.4           | 310           | 350   | 244           | 291              |
| 37/0.093      | 150           | 200           | 1/0           | 1/0           |               |      | 3             | 0.34          | 350           | 405   |               | 333              |
| 37/0.130      | 185           | 250           | 3/0           | 3/0           |               |      | 3.5           | 0.29          | 390           | 460   |               | 381              |
| 61/0.093      | 240           | 300           | 3/0           | 3/0           |               |      | 4             | 0.24          | 450           | 555   |               | 452              |
| 61/0.103      | 300           | 425           | 3/0           | 3/0           |               |      | 4             | 0.22          | 515           | 640   |               | 526              |
| 91/0.093      | 400           | 585           | 3/0           | 3/0           |               |      | 6             | 0.2           | 586           | 770   |               | 639              |
| 91/0.103      | 500           | 685           | 3/0           | 3/0           |               |      | 6             | 0.18          | 680           | 900   |               | 752              |
| 127/0.103     | 630           | 800           | 3/0           | 3/0           |               |      | 6             | 0.17          | 800           | 1030  |               | 855              |



The background of the image is a complex, abstract architectural line drawing. It features a dense arrangement of geometric shapes, primarily rectangular and triangular, which are interconnected to form a three-dimensional structure. The lines are thin and black, set against a plain white background. The perspective is from a high angle, looking down into the structure, which creates a sense of depth and complexity. The overall effect is that of a technical or architectural sketch, possibly representing a modern building or a large-scale infrastructure project.

**THANK YOU**